Duration of antibody protection project

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Abstract

How much longitudinal antibody data must a vaccine trial accumulate before it can make a credible statement about the duration of protection? At a planned review, this becomes a practical design question: is the evidence sufficient for durability follow-up to conclude, or is further observation still likely to change the answer? This workshop project will approach that question through simulation and model-based extrapolation. The working application is COVID-19 vaccination, for which both antibody-kinetics models and quantitative links between neutralisation and protection are available. The choices below are starting points for discussion, not a fixed analysis protocol.

1 Project question and aims

Consider a vaccine trial, or a long-term extension, that measures an antibody marker repeatedly after vaccination. Antibody levels often fall quickly at first and more slowly later. We would like to estimate when the marker will fall below a protection-relevant level, even when that crossing occurs beyond the observed follow-up. Immune correlates require careful terminology—a correlate can predict protection without being its sole cause—but they can still support useful quantitative predictions (Plotkin and Gilbert 2012; Earle et al. 2021).

The central question is therefore:

> *At a planned trial review, how long must participants have been followed, and how informative must their longitudinal measurements be, before the duration of antibody-associated protection can be estimated credibly enough to decide whether further durability follow-up is needed?*

The project has four linked aims:

1. describe plausible short- and long-term antibody trajectories;
2. quantify how much extrapolation is being asked of a fitted model;
3. understand when apparently similar fits imply very different durations of protection;
4. translate those findings into practical guidance on the duration and timing of trial follow-up.

The distinction between concluding durability follow-up and stopping all trial activity is important: safety surveillance, clinical endpoints and regulatory commitments may independently require continued follow-up. The present project concerns the information available for the durability question.

This decision problem is clearest when protection is expected to last for years. The pivotal Shingrix trials followed participants for roughly three to four years, while their long-term extension ultimately assessed efficacy and immunogenicity through 11 years after vaccination (Strezova et al. 2025). Such studies show the value of long follow-up, but also motivate asking at which earlier review points the accumulated marker data can already support a reliable conclusion, and when additional observations are still needed to resolve the tail.

COVID-19 provides a well-developed starting point: neutralisation is predictive of protection, and published work connects titres, variants, boosting and vaccine effectiveness (Khoury et al. 2021; Cromer et al. 2022; Gilbert et al. 2022; Feng et al. 2021). Whether the same framework transfers to other pathogens, assays and types of immune marker is deliberately left open for the workshop.

2 What do we mean by duration of protection?

Let $c_i(t)$ denote the latent antibody concentration for participant i at time t, and let c_{thr} be a protection-relevant threshold on the same assay scale. A natural individual-level target is the first threshold-crossing time

$$T_i^* = \inf\{t \geq 0 : c_i(t) \leq c_{\text{thr}}\}. \tag{1}$$

This quantity is a model-based estimand, not a directly observed biological expiry date. It can be undefined if the trajectory approaches a plateau above the threshold, and it depends on the endpoint and threshold definition.

For the initial COVID-19 application, let c_{conv} denote the mean convalescent neutralisation level and express both $c_i(t)$ and c_{thr} relative to that reference. One option is then to anchor c_{thr} to the level associated with 50% protection in the logistic relationship estimated by Khoury et al. (2021). The same fitted antibody curve could instead be passed through the full titre-to-protection relationship to obtain a protection curve $E_i(t)$. This makes explicit that "antibody duration" and "protection duration" are related but not interchangeable.

Heterogeneity creates a second distinction. The distribution of individual T_i^* values is not equivalent to the threshold-crossing time of a population-average antibody curve. A population target might instead be the time at which the fraction of people still above the threshold falls below a chosen level. The workshop should keep individual and population questions separate.

3 Candidate trajectory models

The candidate set is intended to expose different assumptions about the unobserved tail. We suggest the biphasic exponential as a common reference model for the first analyses, while retaining the other families as substantive alternatives.

Single exponential

$$c(t) = c_0 e^{-kt}. \tag{2}$$

This is the simplest reference model. The log concentration has a constant slope and $T^* = k^{-1}\log(c_0/c_{\text{thr}})$ when $c_0 > c_{\text{thr}}$. It can be identified from relatively short follow-up if the trajectory is truly exponential, but it extrapolates the early slope indefinitely when that assumption is wrong. Single-half-life summaries are common in analyses of COVID-19 neutralisation waning (Khoury et al. 2021; Cromer et al. 2022).

Biphasic exponential (suggested default)

Biphasic structures have been used successfully to describe vaccine-antibody kinetics in established analyses, representing rapid initial contraction followed by a slower persistent component (M. T. White et al. 2015; Hogan et al. 2023). We therefore suggest this family as the initial reference model: its parameters have a useful biological interpretation, its long-term driver is explicit, and a published COVID-19 calibration is available. This is a practical starting point for comparison, rather than a claim that biphasic decay will be preferred for every dataset. We use the smooth parameterisation of Hogan et al. (2023) from the outset:

$$c(t) = c_0\{w_1 e^{-k_1 t} + w_2 e^{-k_2 t}\}, \qquad k_r = \frac{\log 2}{h_r}, \quad r \in \{1, 2\}, \quad h_1 < h_2, \tag{3}$$

$$D = e^{-k_1 t_s} + e^{-k_2 t_s}, \qquad w_1 = \frac{e^{-k_2 t_s}}{D}, \qquad w_2 = \frac{e^{-k_1 t_s}}{D}. \tag{4}$$

The fast component can represent contraction of short-lived antibody-secreting cells and the slow component a longer-lived population. The weights ensure $c(0) = c_0$ and make the two weighted components equal at switching time t_s. The slow half-life h_2 governs long-term duration but may be weakly identified while the fast component dominates.

Power-law decay

$$c(t) = c_0(1 + \alpha t)^{-\beta}. \tag{5}$$

This curve decelerates continuously. It can resemble an exponential over a short interval while implying a much longer tail. Long-term antibody data provide biological motivation for considering decelerating decay rather than a constant half-life (Amanna et al. 2007).

Plateau or set-point

$$c(t) = A_\infty + (c_0 - A_\infty)e^{-kt}. \tag{6}$$

Here the long-term level is non-zero. A crossing exists only when $A_\infty < c_{\text{thr}}$. Short data may not distinguish a genuine plateau from a very slow decline, making "never crosses" an informative possibility rather than a numerical failure. Long-term measurements showing durable antibody levels motivate this alternative (Amanna et al. 2007).

Mechanistic antibody-production model

A richer generating model can represent short- and long-lived antibody-secreting cells and antibody clearance separately:

$$\frac{dP_s}{dt} = -\delta_s P_s, \qquad \frac{dP_l}{dt} = -\delta_l P_l, \qquad \frac{dc}{dt} = \rho_s P_s + \rho_l P_l - \delta_c c. \tag{7}$$

When antibody clears quickly relative to the cells, this approaches biphasic decay; added memory or replenishment can generate an even slower tail. Related models have been developed for plasma cell dynamics and vaccine responses (Andraud et al. 2012; Pasin et al. 2019). Such a model may be most useful as a deliberately richer data-generating mechanism rather than as the default fitted model.

A complementary later-stage generator could use the semi-mechanistic model developed across a series of COVID-19 studies (Nishiyama et al. 2023; Nakamura et al. 2024; Park et al. 2025; Hart et al. 2025). It links a decaying mRNA stimulus to delayed, nonlinear antibody production and

antibody loss, and can represent boosting, infection history and between-participant variation. Public code and derived parameter estimates make it a practical calibration resource. Once vaccine stimulation has vanished, however, its long-term tail tends towards a single exponential, so it is better suited to testing dose-response dynamics and heterogeneous trajectories than to replacing the richer slow-tail generator above.

4 Observation model and biphasic population parameterisation

Observed assays are naturally treated as noisy measurements of a latent positive curve. A simple starting point is multiplicative log-normal error,

$$\log y_{ij} = \log c_i(t_{ij}) + \epsilon_{ij}, \qquad \epsilon_{ij} \sim N(0, \sigma^2_{\log}). \tag{8}$$

Lower and upper quantification limits can be represented as censoring rather than replacing censored observations by exact boundary values. This observation model can be used with any of the candidate trajectories in Section 3.

For the suggested biphasic reference model, between-participant variation can be introduced through correlated log-normal random effects on peak level and the slow decay rate, for example

$$c_{0i} = c_0 e^{\eta_{0i}}, \qquad k_{2i} = k_2 e^{\eta_{ki}}, \tag{9}$$

$$\begin{pmatrix} \eta_{0i} \\ \eta_{ki} \end{pmatrix} \sim N_2 \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \omega_0^2 & \rho_{0k}\omega_0\omega_k \\ \rho_{0k}\omega_0\omega_k & \omega_k^2 \end{pmatrix} \right]. \tag{10}$$

This permits both individual and population targets and asks whether additional participants compensate for sparse information within each trajectory. Population mechanistic models have used this type of structure in vaccine applications (Pasin et al. 2019).

Table 1 makes the first case study concrete by using Hogan's estimate for BNT162b2 after dose 3 against Delta, alongside their shared biphasic decay parameters. All antibody levels use the convalescent-relative scale defined above. These values are calibration aids, not universal biological constants; the source/basis column distinguishes published estimates from workshop modelling choices.

Table 1: Tentative starting values for a BNT162b2 dose-3 biphasic COVID-19 baseline.

Quantity	Starting value	Source / basis	Interpretation / possible variation
Early level c_0	$0.92c_{\text{conv}}$	Hogan et al. (2023)	BNT162b2 dose 3 against Delta; 95% CrI 0.71–1.19
Log-error SD $\sigma_{\log}$	0.20	Workshop noise choice	Explore lower and higher assay variability
Fast half-life h_1	35 days	Hogan et al. (2023)	Early contraction; tentative range 25–60 days
Slow half-life h_2	581 days	Hogan et al. (2023)	Highest-priority tail sensitivity; consider 250–900 days
Switching time t_s	75 days	Hogan et al. (2023)	Equal weighted components at this time
Weights $w_1 : w_2$	about 80:20	Derived from Hogan et al. (2023)	From h_1, h_2 and t_s in Equation 4
Protection anchor c_{thr}	0.202 / 0.030	Khoury et al. (2021)	Multiples of c_{conv} for 50% symptomatic / severe protection
Peak / rate log-SD ω_0/ω_k	0.30 / 0.25	Workshop heterogeneity choice	Moderate between-participant variation
Peak–rate correlation ρ_{0k}	-0.30	Workshop correlation choice	Include zero as a comparison

The uncited workshop choices in the table are deliberately round simulation defaults, not literature estimates. The workshop may choose a different vaccine, dose, assay normalisation or tail calibration before any headline analysis.

5 Proposed priors with a common early decline

For the initial homogeneous analysis, we propose matching prior information about the early antibody trajectory across candidate families. Their long-term predictions may differ: an exponential can estimate early decline precisely yet confidently misestimate a much later threshold crossing. The priors below are workshop modelling choices centred on the calibration in Table 1, not published estimates of prior uncertainty. They specify fitting assumptions separately from the generating truth.

Fit each family once at each review, then calculate both crossing times from every posterior trajectory draw, using $c_{\text{sym}} = 0.202$ and $c_{\text{sev}} = 0.030$. This preserves their joint uncertainty and gives $T^*_{\text{sym}} \leq T^*_{\text{sev}}$. Keep the selected prior specification fixed across review dates, cohort sizes and truth scenarios; do not recalculate its centres from each simulated dataset. The thresholds are fixed protection anchors for this initial comparison, with their uncertainty reserved for a separate sensitivity analysis.

Common early-trajectory parameters

Let the reference day be $t_a = 90$ and define the positive cumulative log decline and the fraction remaining by

$$D_a = -\log\{c(t_a)/c_0\} > 0, \qquad R_a = e^{-D_a}. \tag{11}$$

Every family is parameterised so that $c(0) = c_0$ and $c(t_a) = c_0 R_a$. Consequently, the joint prior for these two titres is identical across families. Curvature between these times remains model-specific. The reference day stays at 90 days even when a review occurs earlier or later; it defines a parameter, not an additional observation.

Consistent model parameterisations

The following transformations introduce the additional free variables and ensure that every family has the common initial titre and day-90 decline. All logarithms are natural, and all times are in days.

Single exponential. No additional free parameter is needed. Set

$$k = D_a/t_a, \qquad c(t) = c_0 e^{-kt}. \tag{12}$$

The model extends this early average log decline indefinitely; its late slope is not adjusted to resemble the more flexible families.

Biphasic exponential. Introduce $u_f \in (0,1)$ and $z_q \in \mathbb{R}$. Set $f = u_f/2$, $q = 1 + e^{z_q}$, and write

$$c(t) = c_0\{(1-f)e^{-qst} + fe^{-st}\}, \qquad s = k_2, \quad q = k_1/k_2. \tag{13}$$

Here f is the fraction of the initial titre contributed by the slow component, rather than a fraction of participants. Given D_a, f, q, obtain $s > 0$ as the unique solution of

$$(1-f)e^{-qst_a} + fe^{-st_a} = R_a. \tag{14}$$

The left-hand side decreases continuously from 1 to 0, so a solution exists for every allowed draw. This is the same trajectory family as Equation 4, with

$$h_1 = \frac{\log 2}{qs}, \qquad h_2 = \frac{\log 2}{s}, \qquad t_s = \frac{\log\{(1-f)/f\}}{(q-1)s}. \tag{15}$$

The restriction $0 < f < 0.5$ specifies a positive switching time and an initially larger fast contribution. The half-lives, switching time and slow rate are derived from D_a, u_f, z_q rather than free parameters.

Power-law decay. Introduce the free exponent $\beta > 0$. Given D_a and β, set

$$\alpha = \frac{e^{D_a/\beta} - 1}{t_a}, \qquad c(t) = c_0(1+\alpha t)^{-\beta}. \tag{16}$$

The exponent controls deceleration while α is determined by the common day-90 decline and β.

Plateau or set-point. Introduce $u_\infty \in (0,1)$, the floor as a fraction of the day-90 titre, and set

$$A_\infty = u_\infty c_0 R_a, \qquad r = A_\infty/c_0, \tag{17}$$

$$k = -\frac{1}{t_a}\log\left(\frac{R_a - r}{1-r}\right), \qquad c(t) = A_\infty + (c_0 - A_\infty)e^{-kt}. \tag{18}$$

Because $0 < r < R_a < 1$, the rate is positive and the floor lies below the day-90 titre. Depending on the floor, a trajectory can cross both thresholds, only the higher threshold, or neither.

Default and broader prior options

With these parameterisations, Table 2 specifies priors on the common variables $\log c_0, \log D_a, \log \sigma_{\log}$ and each family's additional free variables. All listed variables are mutually independent within a family; derived rates, half-lives and titres will generally be dependent. Here $N(\mu, s^2)$ denotes a normal distribution with standard deviation s.

Table 2: Proposed fitting priors for the homogeneous comparison. Choose a whole column for a run; the common priors apply to every candidate family.

Free variable	Default	Broader alternative
Common: $\log c_0$	$N(\log 0.92,\ 0.5^2)$	$N(\log 0.92,\ 0.8^2)$
Common: $\log D_a$	$N(\log 1.16,\ 0.5^2)$	$N(\log 1.16,\ 0.8^2)$
Common: $\log \sigma_{\log}$	$N(\log 0.20,\ 0.5^2)$	$N(\log 0.20,\ 0.7^2)$
Biphasic: u_f	$\mathrm{Beta}(2,3)$	$\mathrm{Beta}(1,1.5)$
Biphasic: z_q	$N(\log(581/35-1),\ 0.8^2)$	$N(\log(581/35-1),\ 1.1^2)$
Power law: $\log \beta$	$N(\log 1.5,\ 0.7^2)$	$N(\log 1.5,\ 1.0^2)$
Plateau: u_∞	$\mathrm{Beta}(1,1)$	$\mathrm{Beta}(1,1)$

Sample these free variables and transform them as above; do not add independent priors on derived quantities such as h_1, h_2, t_s, s or α.

The centre $D_a = 1.16$ is the rounded day-90 log decline of the reference biphasic curve. Under the default, the approximate 95% prior ranges are 0.35–2.45 for c_0, 0.045–0.65 for R_a, and 0.075–0.53 for $\sigma_{\log}$. The broader option widens these to 0.19–4.41, 0.0038–0.79, and 0.051–0.79, respectively. It retains the same log-scale centres and the same mean $f = 0.20$, while reducing concentration

around them. The plateau fraction is already uniform and is left unchanged. "Broader" refers to these sampling scales, not to a claim of less information about every derived quantity: more diffuse priors can increase mass on extremely long durations or initial titres below a threshold.

For the exponential, the default median half-life is approximately 54 days. For the biphasic model, f has mean 0.20 and an approximate 95% range of 0.034–0.403; q has median 16.6 and an approximate 95% range of 4.3–75.8. At $D_a = 1.16$, $f = 0.20$ and $q = 581/35$, the implied half-lives are about 35 and 579 days, with a 74-day switching time. For the power law, the default approximate 95% range for β is 0.38–5.91; the broader option gives 0.21–10.65.

For the plateau, the uniform prior on u_∞ permits all three crossing outcomes. Under the default common priors, the prior non-crossing probabilities are approximately 0.29 for the symptomatic anchor and 0.83 for the severe anchor. These are prior implications of the floor assumption and should be reported alongside posterior probabilities, rather than attributed entirely to the data.

Interpretation and checks before scored runs

Table 3 illustrates the intended agreement in early behaviour and disagreement in extrapolation. It uses selected free parameter values, not marginal prior medians of the crossing times. Its homogeneous trajectory has one common duration for all participants; these values do not represent a heterogeneous population-duration estimand.

Table 3: Illustrative curves at $c_0 = 0.92$, $D_a = 1.16$, and, where applicable, $f = 0.20$, $q = 581/35$, $\beta = 1.5$, $u_\infty = 0.5$. Durations are rounded to whole days.

Family	$c(30)$	$c(90)$	T^*_{sym} (days)	T^*_{sev} (days)
Exponential	0.625	0.288	118	266
Biphasic	0.583	0.288	139	1516
Power law	0.562	0.288	135	678
Plateau	0.587	0.288	139	∞

Before scored runs, inspect prior predictive trajectories over both the observed and extrapolated periods, including the joint distribution of the two crossing times and probabilities of remaining above each threshold at fixed horizons. Retain genuinely non-crossing draws as $T^* = \infty$; a crossing beyond a numerical search horizon is a different event. A draw starting at or below a threshold has crossing time zero. Compare the default and broader columns as declared sensitivity runs, reporting whether a short-follow-up severe-duration conclusion depends on the tail priors. These priors specify homogeneous fits only; the hierarchical extension needs an explicit joint specification for population and random-effect parameters.

An alternative with common day-90 and day-365 anchors

As a separate sensitivity formulation, match the joint prior on $c_0, c(90), c(365)$ across the three flexible families. Retain the original initial-level and day-90 prior for the exponential, which still extrapolates its early decline unchanged. This supplies common information about the first year without requiring agreement about later threshold crossings. The anchor days define latent parameters, not additional observations; their priors remain fixed at every review.

Set $t_b = 365$, $m = t_b/t_a$, and introduce

$$\rho = \frac{D_b - D_a}{(m-1)D_a} \in (0,1), \qquad D_b = D_a\{1 + (m-1)\rho\}, \qquad R_b = e^{-D_b}. \tag{19}$$

Thus ρ is the average log-decay rate from day 90 to 365 divided by that from day 0 to 90. It enforces $R_a^m < R_b < R_a$: positive decline with deceleration. The reference biphasic trajectory gives $\rho \simeq 0.25$. Independent priors directly on the three titres would not enforce these constraints.

Power law. Let $v > 0$ solve

$$\frac{\log\{1 + m(e^v - 1)\}}{v} = \frac{D_b}{D_a}, \qquad \beta = D_a/v, \qquad \alpha = (e^v - 1)/t_a. \tag{20}$$

The ratio decreases from m to 1, so the anchors determine the two shape parameters uniquely. Use the power-law curve above.

Biphasic exponential. Introduce one additional free variable $u_s \in (0, 1)$ and set

$$k_2 = u_s \rho D_a / t_a, \qquad x = e^{-k_2 t_a}. \tag{21}$$

The slow rate is a fraction of the average day-90-to-365 log-decay rate. Find $y \in (0, R_a)$ as the unique solution of

$$f = \frac{R_a - y}{x - y}, \qquad (1 - f)y^m + fx^m = R_b, \qquad k_1 = -\log(y)/t_a. \tag{22}$$

The left-hand side decreases from $R_a x^{m-1} > R_b$ to $R_a^m < R_b$ as y runs from 0 to R_a. Every draw therefore gives positive ordered rates and $0 < f < 1$, without rejecting or reweighting the common anchors. Use Equation 13 with $s = k_2$ and $q = k_1/k_2$; f, both half-lives and t_s are derived. The reference curve has $u_s \simeq 0.37$.

For this formulation, relax the earlier restriction $f < 0.5$. Some valid anchor combinations require a larger initial slow contribution; their equal-contribution time t_s is negative. Restricting f independently and rejecting incompatible draws would change the anchor distribution. This is an explicit difference between the two prior formulations.

Plateau. Use Equation 22 with $x = 1$, representing a zero-rate slow component. Set $A_\infty = c_0 f$ and $k = -\log(y)/t_a$ in the plateau curve. Both shape parameters are then determined by the anchors. A positive floor still permits infinite crossing times.

Table 4: Two-anchor sensitivity priors. The initial-level, day-90 decline and observation-error priors retain their respective columns in Table 2. These replace the model-specific shape priors in that table, rather than supplementing them.

Additional free variable	Default	Broader alternative
Flexible families: $\text{logit}\,\rho$	$N(\text{logit}\,0.25,\ 1.0^2)$	$N(\text{logit}\,0.25,\ 1.5^2)$
Biphasic only: u_s	$\text{Beta}(2, 3)$	$\text{Beta}(1, 1.5)$

All free variables in this formulation are independent. The two priors on ρ have approximate 95% ranges 0.045–0.70 and 0.017–0.86; both priors on u_s have mean 0.40. These are workshop choices, not external estimates. Neither the earlier f, q priors nor the earlier β, u_∞ priors are also imposed. In particular, the plateau's non-crossing probabilities must be recomputed under these priors.

Inspect prior trajectories and both crossing distributions before fitting, including the implied distribution of f and probability $f > 0.5$. Agreement at three anchors does not imply agreement between them or in the tail. Matching at day 365 also supplies prior information about an unobserved period at early reviews, so compare this formulation with the original early-only formulation rather than treating it as neutral.

6 A possible analysis programme

A simulation-first programme would create a full longitudinal dataset from a known trajectory, then reveal the portions available at successive planned trial reviews. Each candidate model could be fitted to the data available at each review and extrapolated to T^*. Because the generating truth is known, the exercise can separate uncertainty from confident misspecification and ask when a decision to conclude durability follow-up would have been well supported.

A natural progression is:

1. establish a transparent homogeneous baseline in which a biphasic model is fitted to biphasic data;
2. reveal successively longer follow-up windows and examine when conclusions about duration become stable, or only appear stable;
3. fit all candidate families to the same truncated data to expose disagreement in the tail;
4. generate data from richer mechanistic models and ask whether simple fitted models remain useful;
5. introduce participant heterogeneity and compare individual with population duration;
6. if useful, map antibody trajectories to symptomatic and severe-disease protection or compare the framework with an open real dataset.

Points for discussion include which duration is clinically meaningful; how strongly to privilege COVID-19; which model families and parameterisations are worth pursuing; how inference and study design should be handled; what evidence should be sufficient to conclude durability follow-up; which comparisons would be most informative; and what degree of extrapolation would be acceptable to a trial designer.

A single-participant computational pilot

An initial illustration of the first three steps uses one homogeneous participant and one simulated history from Table 1. Monthly measurements include day zero and visits at rounded days $365.25j/12$, revealed at the configured 3, 6, 9, 12, 18 and 24 month reviews. Both prior columns in Table 2 are fitted to exactly the same measurements. Repeat with both columns of Table 4, using exactly the same simulated history, to separate differences due to the prior formulation from observation noise. This single realisation describes individual posterior uncertainty and misspecification; it cannot estimate empirical coverage or identify an adequate trial design.

Use four independent JAGS chains, initially with 2000 adaptation iterations, 2000 further warm-up iterations and 4000 retained draws per chain. Require rank-normalised split $\widehat{R} < 1.01$ and bulk and tail effective sample sizes of at least 400 for the free parameters and duration-relevant functions. For a potentially infinite duration, diagnose the bounded transform $T^*/(365.25 + T^*)$, defined as 1 at infinity, and separately check a non-constant non-crossing indicator. If a duration quantile falls on an atom so that its quantile indicator is constant, its tail effective sample size is not applicable; use the non-constant quantile indicators and the bulk diagnostic. Constant indicators are reported as such rather than assigned a fictitious effective sample size. Extend sampling if needed, recording every attempt and unresolved failure.

Compare posterior trajectories and both duration distributions. As a numerical check, calculate extreme power-law prior durations on the log scale before taking quantiles. A finite duration beyond floating-point range is not a non-crossing trajectory; record a numerical failure if it prevents

a fitted result from meeting the shared output contract, retaining the sampler output for inspection. As a secondary descriptive comparison, report pointwise PSIS leave-one-out expected log predictive density for the observed log titres, with its Pareto-k diagnostics; values above 0.7 flag an unreliable approximation. This comparison concerns interpolation among observed visits and does not validate long-term extrapolation. Also evaluate predictions for the subsequent observed visits withheld at each review, clearly separating this simulation check from information available to the fit. Inspect prior predictive trajectories under both prior sets before interpreting results.

Repeated single-participant estimation study

For the first repeated-simulation comparison, use the early-only default column of Table 2. Generate $R = 100$ independent noisy histories of one participant each, holding the biphasic generating parameters, monthly visits, noise and six reviews fixed as above. Each history is shared across models and nested reviews. Fit all four families with the same sampler and convergence rules, giving 2400 planned fits. This assesses repeated-measurement uncertainty at one fixed truth, not between-participant variation or prior-predictive calibration. Derive every simulation and chain seed from the design and replicate.

For endpoint e and successful replicate r, let m_{re} be the posterior median and $[L_{re}, U_{re}]$ its central 90% interval. Report

$$\widehat{C}_e = \frac{1}{R_{\text{ok}}} \sum_r \mathbf{1}\{L_{re} \le T_e^* \le U_{re}\}, \qquad \widehat{B}_e = \frac{1}{R_{\text{ok}}} \sum_r (m_{re} - T_e^*). \tag{23}$$

Keep infinite medians and interval endpoints. Summarise precision by the median interval width, taking an interval with either endpoint infinite to be unbounded for this summary, and separately report the unbounded interval fraction, infinite-median fraction and mean posterior non-crossing probability. Do not report finite-only bias as unconditional bias.

For a finite observed truth y, CRPS has the CDF representation (Gneiting and Raftery 2007)

$$\text{CRPS}(F, y) = \int_0^\infty \{F(t) - \mathbf{1}(y \le t)\}^2 \, dt. \tag{24}$$

It diverges when F has positive non-crossing mass, and may also diverge for heavy finite tails. The primary comparable score is therefore CRPS for $Z = g(T^*)$, where $g(t) = t/(\tau + t)$, $g(\infty) = 1$ and $\tau = 365.25$ days. This is a one-to-one bounded transformation, so its CRPS scores the full extended duration distribution. Also report the restricted day-scale score

$$\text{CRPS}_H(F, y) = \int_0^H \{F(t) - \mathbf{1}(y \le t)\}^2 \, dt, \qquad H = 3652.5 \text{ days}, \tag{25}$$

equivalently CRPS of $\min(T^*, H)$ against $\min(y, H)$. Check horizons 1826.25 and 7305 days as sensitivities. These caps apply only to the score, not to posterior crossing times. Lower scores are better. Compute empirical CDF integrals exactly between sorted draws, avoiding cancellation of large terms in the pairwise-distance formula. Retain empirical unrestricted day-scale CRPS as a tail diagnostic; a finite Monte Carlo value does not establish finiteness of the underlying score.

Also report mean bias and mean interval width on the bounded g scale, which remain defined when raw duration summaries are infinite. Give Monte Carlo standard errors for finite means and Wilson 95% intervals for coverage and proportions. Report planned, completed and converged counts for every model/review; primary metric summaries use converged fits only. Bound all-planned coverage by assigning every unsuccessful fit first a coverage failure and then a success. Retain unresolved fits and numerical failures in the outputs. Pointwise LOO is not needed for this study.

7 How the literature informs the project

Plotkin and Gilbert (2012). This short framework gives us the vocabulary needed to distinguish a statistical correlate, a mechanistic correlate and a surrogate endpoint, helping us avoid interpreting a protection-relevant antibody marker as the whole immune mechanism (Plotkin and Gilbert 2012).

Khoury et al. (2021). This is the main quantitative bridge from neutralisation titre to protection: it supplies the logistic relationship and relative 50%-protection anchors that can be used to turn a fitted antibody curve into an endpoint-specific protection curve (Khoury et al. 2021).

Cromer et al. (2022). The meta-analysis extends the titre-to-protection framework across variants and boosting, reminding us that starting titre and antigenic change can alter protection duration even when the underlying decay process is unchanged (Cromer et al. 2022).

Gilbert et al. (2022). The mRNA-1273 efficacy-trial analysis provides trial-level evidence that neutralisation is a graded correlate of risk, supporting continuous marker models rather than only a binary laboratory cut-off (Gilbert et al. 2022).

Feng et al. (2021). This analysis of ChAdOx1 nCoV-19 similarly supports a continuous relationship between immune markers and symptomatic or asymptomatic infection, and cautions against treating any single threshold as universal (Feng et al. 2021).

Earle et al. (2021). The cross-vaccine synthesis supplies broader evidence that antibody levels can function as a protective correlate for COVID-19 vaccines, strengthening the case for the marker-based starting point while not establishing causality by itself (Earle et al. 2021).

Hogan et al. (2023). This model-based study provides vaccine- and dose-specific initial immunity levels and the tentative COVID-19 biphasic calibration—including fast and slow half-lives and a switching time—and illustrates how immune kinetics can be connected to longer-term effectiveness against variants (Hogan et al. 2023).

Nishiyama et al. (2023). This study extends an mRNA-stimulus antibody model to a booster dose and infection history, showing how repeated immune exposures can alter response duration and providing a compact semi-mechanistic alternative for later simulations (Nishiyama et al. 2023).

Nakamura et al. (2024). Dense measurements from a small calibration cohort and sparse measurements from more than 2,400 participants are combined to reconstruct heterogeneous individual trajectories after primary vaccination. The fitted parameter distributions are a useful COVID-19 calibration source, although the underlying clinical dataset is access-restricted (Nakamura et al. 2024).

Park et al. (2025). Longitudinal trajectories after primary and booster vaccination are stratified into durable, vulnerable and rapid-declining response patterns, with rapid decline linked to earlier subsequent infection. This makes heterogeneity clinically relevant rather than merely a statistical complication (Park et al. 2025).

Hart et al. (2025). The same individual-level antibody model is linked to susceptibility and population outbreak risk. Its open code and derived parameter estimates provide an immediately usable resource for simulation and calibration (Hart et al. 2025).

White et al. (2015). The RTS,S malaria analysis is the closest published analogue to the project's intended chain: it uses biphasic antibody kinetics and links the resulting waning marker to duration of vaccine efficacy (M. T. White et al. 2015).

Strezova et al. (2025). The final Shingrix long-term follow-up analysis demonstrates the practical scale of the design problem: an extension of pivotal phase 3 trials followed more than 7,000 participants and assessed protection through 11 years. It provides a concrete setting in which the value of additional durability data must be weighed against prolonged follow-up (Strezova

et al. 2025).

Andraud et al. (2012). The nested plasma-cell and antibody models provide a clear mechanistic template for generating trajectories on several biological time scales and for asking when a simpler decay curve is an adequate approximation (Andraud et al. 2012).

Pasin et al. (2019). The Ebola prime-boost study demonstrates how mechanistic antibody dynamics and between-participant variation can be estimated together, making it a useful model for the hierarchical phase of this project (Pasin et al. 2019).

Amanna et al. (2007). Long-term measurements across antigens show that antibody durability varies greatly and need not follow a single constant half-life, motivating the power-law and plateau alternatives and explicit attention to the tail (Amanna et al. 2007).

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